ERGON Foundation

Amazon Lightsail Hosting & Upload Storage Guide
Reports, gallery images, gallery videos and admin management

1. Correct Lightsail setup

Users → Domain + HTTPS → Lightsail Instance (Nginx + Node.js + React build)
↘ PostgreSQL + persistent uploads folder on an attached Lightsail disk

Component	Where it runs
Website and admin panel	Same Lightsail Linux instance, served by Nginx.
Node.js API	Same instance, started by PM2 or Docker Compose.
PostgreSQL	Prefer a Lightsail managed database; for a small initial setup, Docker/PostgreSQL on the instance is possible but must be backed up.
Reports and gallery uploads	A mounted Lightsail block-storage disk, not the temporary container filesystem.

Important: do not keep uploaded files only inside a Docker container. A redeploy or replacement can remove them. Store uploads on the attached disk and mount that same folder into the backend container.

2. Create persistent upload storage

1. In Lightsail, choose **Storage** → **Create disk**.
2. Create a disk in the **same Region and Availability Zone** as the ERGON instance. Name it `ergon-uploads`.
3. Attach it to the ERGON instance.
4. Connect to the instance over SSH and identify the disk: `lsblk`.
5. For a new empty disk, format it once (confirm the device name first): `sudo mkfs.ext4 /dev/nvme1n1`.
6. Mount it at `/mnt/ergon-data` and persist the mount in `/etc/fstab`.
7. Create the application upload folders: `sudo mkdir -p /mnt/ergon-data/uploads/{reports,gallery/images,gallery/videos}`.

```
sudo mkdir -p /mnt/ergon-data
sudo mount /dev/nvme1n1 /mnt/ergon-data
sudo mkdir -p /mnt/ergon-data/uploads/reports
sudo mkdir -p /mnt/ergon-data/uploads/gallery/images
sudo mkdir -p /mnt/ergon-data/uploads/gallery/videos
sudo chown -R ubuntu:ubuntu /mnt/ergon-data/uploads
```

```
# Add the UUID from: sudo blkid /dev/nvme1n1
# UUID=<disk-uuid> /mnt/ergon-data ext4 defaults,nofail 0 2
```

3. Connect the project to the disk

When using Docker Compose

Mount the disk folder into the API container. The container must see it as `/app/uploads`, or set the application upload path to the mounted location.

```
services:
  api:
    volumes:
      - /mnt/ergon-data/uploads:/app/uploads
```

When running Node.js directly with PM2

Set the application upload directory to `/mnt/ergon-data/uploads`, or create a symlink from the project folder:

```
ln -s /mnt/ergon-data/uploads /var/www/ergon-foundation/uploads
```

4. Required Nginx setting

Users must be able to read uploaded files. Serve the persistent folder directly through `/uploads/`.

```
location /uploads/ {
  alias /mnt/ergon-data/uploads/;
  try_files $uri =404;
  add_header X-Content-Type-Options nosniff;
}
```

Then test a real file: `https://your-domain.com/uploads/gallery/images/filename.png`. It must return the image without login.

5. How the upload process works

1. Admin uploads a report or gallery file.
2. The API validates its type and size, stores it on `/mnt/ergon-data/uploads`, and saves its relative path in PostgreSQL.
3. For example: `gallery/images/unique-file.png` or `reports/unique-file.pdf`.
4. The public Gallery and Reports APIs return the saved records.
5. The public website loads the file at `/uploads/<saved-path>`.

Content	Accepted formats	Current maximum
Reports	PDF, JPG/JPEG, PNG, GIF, WEBP	20 MB
Gallery images	JPG/JPEG, PNG, GIF, WEBP	10 MB
Gallery videos	MP4, WEBM	100 MB

6. Deployment process

1. Before deployment, confirm `/mnt/ergon-data/uploads` is mounted: `df -h`.
2. Deploy new application code only. Do not delete the mounted uploads folder.
3. Restart API/PM2/Docker Compose.
4. Test admin upload, API listing, and public URL.
5. After a successful test, take a manual Lightsail snapshot before major upgrades.

7. Backup and recovery

- Enable Lightsail automatic snapshots for the instance and the attached disk. Lightsail retains the latest seven automatic snapshots.
- Create a manual snapshot before operating-system upgrades, database migrations, or major releases.
- If PostgreSQL is local, make a daily compressed `pg_dump` into the attached disk, for example `/mnt/ergon-data/backups`.
- Periodically download or copy a backup to a second location. Test restoring a snapshot before relying on it.

> **Recovery:** create a new Lightsail instance from the instance snapshot; if needed, create and attach a replacement disk from the disk snapshot; mount it at `/mnt/ergon-data`; then deploy the application and restore PostgreSQL.

8. Security checklist

- Allow only ports 80, 443 and SSH (22, restricted to your IP) in the Lightsail firewall.
- Use HTTPS with a valid certificate; configure `CORS_ORIGIN` to the real website domain.
- Keep `DATABASE_URL` and JWT secrets in a server-only `.env` file—never in React code or Git.
- Validate file type and size on the backend. Keep unique generated file names.
- Use a non-root application user and correct ownership for the uploads folder.

Official AWS references

[Create and attach Lightsail block-storage disks]
[Lightsail snapshots and recovery]
[Configure Lightsail automatic snapshots]